

Curriculum Vitae – Sean Xu

Electronic mail: seanjx@berkeley.edu
Mobile phone: +1 509 578 0421

Interests

Cosmology, new probes of cosmology, computational physics, Bayesian inference, tests of fundamental physics, amateur philosophy and political economy.

Education

- B.A. Physics, University of California, Berkeley, expected 2027

Research

- *Gravitational lensing*, 2024–. Projects in Bayesian modeling of galaxy-scale strong gravitational lenses, modeling of the Carousel cluster lens for cosmology, and implementing new inference algorithms with high-performance computing. PI: Xiaosheng Huang (U. San Francisco/LBNL). Partly conducted through a DOE Science Undergraduate Laboratory Internship at Lawrence Berkeley National Laboratory in summers of 2024 and 2026.
- *Nitrogen vacancy centers in diamond*, 2024–. Projects in the construction of magnetic coils, the construction of a laser injection lock, a search for coincident nuclear transitions for heteronuclear Mössbauer spectroscopy, and part fabrication for new NV experiments. PI: Shimon Kolkowitz (UC Berkeley).
- *Random matrix theory and quantum gravity*, 2025–2026. Monte Carlo simulation of random matrix models of quantum gravity, under the auspices of the Physics Directed Reading Program. Mentor: Samuel Weiss (UC Berkeley).

Preprints

- Huang, X., Upson, L., Ratier-Werbin, N., Lu, H., **Xu, S.**, et al. 2026, GIGA-Lens 2.0: Strong-Lens Modeling on Multiple GPU Nodes, <https://arxiv.org/abs/2606.30633>.
- Urcelay, F., Huang, X., Sheu, W., O'Donnell, J.H., Jeltama, T., Williams, D., **Xu, S.**, et al. 2026, The Carousel Lens II: Cosmological Constraints with GIGA-Lens, <https://arxiv.org/abs/2602.16077>.
- Baltasar, S., Ratier-Werbin, N., Huang, X., ..., **Xu, S.** 2026, A Novel Lensed Point Source Modeling Pipeline using GIGA-Lens with Application to SN Zwicky and SN iPTF16geu, <https://arxiv.org/abs/2601.18787>.
- Huang, X., Alvarez-Garcia, D., Ubeda, M., Bhamre, V., **Xu, S.**, et al. 2025, DESI Strong Lens Foundry V: A Sample of HST-Observed Strong Lenses Modeled with GIGA-Lens, <https://arxiv.org/abs/2512.07823>.

Honors and awards

- Berkeley Physics Undergraduate Research Scholar (BPURS), Spring 2026
- BPURS, Spring 2025
- National Merit Scholar, 2023

Course positions

- Tutor for Physics 105, Spring 2026 (Ziqiang Qiu)
- Reader for Physics 5A, Spring 2026 (Mike DeWeese)
- Reader for Astron C10, Fall 2025 (Alex Filippenko)
- Reader for Physics 5A, Spring 2025 (Mike DeWeese)

Other professional activities

- Physics Major Course Commiteeee, 2025–
- Berkeley Society of Physics Students (SPS) senior advisor, 2026–
- SPS webmaster, 2025–2026

Talks

- SPS undergraduate seminar, March 2026
“Astrophysical modeling for cosmology”
- SPS undergraduate seminar, October 2025
“Random aspects of NV centers: quantum sensors and atomic physics methods”
- SPS undergraduate seminar, December 2024
“Graviational lens modeling: computational methods for cosmology”

Posters

- APS Global Summit, March 2025
“Statistical convergence in strong lens modeling on multiple GPU nodes”
also presented at: BPURS Poster Session, April 2025
- FiO/LS Undergraduate Symposium, October 2025
“Measurement of the energy of the singlet NV states”
- BPURS Poster Session, April 2025
“Building a laser injection lock amplifier”
- Berkeley Lab WD&E Summer Intern Symposium, August 2024
“Modeling a high redshift gravitational lens at $z = 1.092$ ”